

Theoretical Prediction of Antiferromagnetic Skyrmion Crystal in Janus Monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$

Kaiying Dou, Wenhui Du, Zhonglin He, Ying Dai,* Baibiao Huang, and Yandong Ma*



Cite This: *ACS Nano* 2023, 17, 1144–1152



Read Online

ACCESS |



Metrics & More



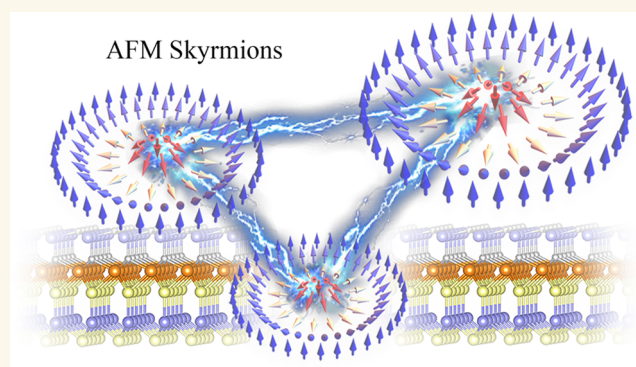
Article Recommendations



Supporting Information

ABSTRACT: An antiferromagnetic skyrmion crystal (AF-SkX), a regular array of antiferromagnetic skyrmions, is a fundamental phenomenon in the field of condensed-matter physics. So far, very few proposals have been made to realize the AF-SkX, and most have been based on three-dimensional (3D) materials. Herein, using first-principles calculations and Monte Carlo simulations, we report the identification of AF-SkX in a two-dimensional lattice of the Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. Arising from the broken inversion symmetry and strong spin–orbit coupling, a large Dzyaloshinskii–Moriya interaction is obtained in the Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. This, combined with the geometric frustration of its triangular lattice, gives rise to the skyrmion physics and long-sought AF-SkX in the presence of an external magnetic field. More intriguingly, this system presents two different antiferromagnetic skyrmion phases, and such a phenomenon is distinct from those reported in 3D systems. Furthermore, by contacting with Sc_2CO_2 , the creation and annihilation of AF-SkX in Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ can be achieved through ferroelectricity. These findings greatly enrich the research on antiferromagnetic skyrmions.

KEYWORDS: antiferromagnetic skyrmion crystal, two-dimensional lattice, first-principles, Dzyaloshinskii–Moriya interaction, Janus monolayer



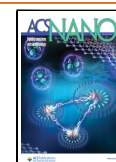
Magnetic skyrmions are localized vortex- or anti-vortex-like spin structures characterized by spins wrapping a unit sphere and protected by an integer topological charge.^{1–4} Their topological properties lead to the exceptional inherent stability against transitions into trivial spin structures, making them technologically promising for data-storage applications.^{5–9} On a fundamental level, magnetic skyrmions emphasize the role of topology in condensed-matter physics.^{3,4} In most systems observed, noncoplanarity in the directions of spins at different lattice sites correlates to a ferromagnetic (FM) exchange interaction: namely, FM skyrmions.^{10–20} Nonetheless, FM skyrmions have shortcomings in common with other spintronics: for example, the sensitivity to stray field.^{3,21} In contrast, topological spin structures in antiferromagnetic (AFM) lattices, consisting of compensated FM sublattices, generate no stray field; moreover, they have been demonstrated to host ultrafast spin dynamics and relativistic motion of textures.^{22–28} Therefore, generating AFM skyrmions is crucial for both fundamental research and applications.^{21,22,29–36}

Despite the huge interest, AFM skyrmions have been experimentally observed only in several thin-film systems, including $\text{Fe}_2\text{O}_3\text{-Pt}$,³⁷ MnSc_2S_4 ,³⁸ and synthetic antiferromagnets.^{39–41} In these few existing systems, isolated AFM skyrmions are usually observed, while the AFM skyrmion crystal (AF-SkX), a regular array of antiferromagnetic skyrmions, is rarely identified.³⁸ With the recent rise of long-range magnetic ordering in two-dimensional (2D) lattices, atomically thin AFM materials have gained a great deal of attention.^{42–45} Physically, triggered by competing exchange interactions, 2D AFM systems can manifest topological spin textures as well.^{30–35} For examples, Rosales et al. explored the existence of AF-SkX in a triangular lattice using a frustrated

Received: August 26, 2022

Accepted: November 23, 2022

Published: November 30, 2022



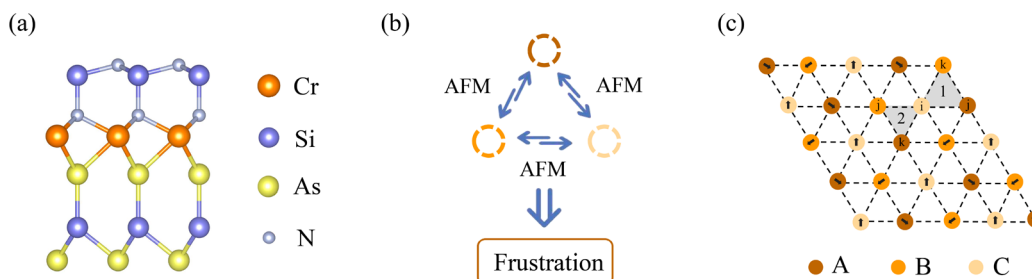


Figure 1. (a) Crystal structure of Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ from a side view. (b) Geometric frustration from antiferromagnetic coupling in a triangular lattice. (c) Diagram of three magnetic sublattices in a triangular lattice.

antiferromagnetic Heisenberg model.³² Mohylina et al. followed up on the Rosales' work³² and investigated the formation and stability of AF-SkX in a triangular lattice.^{46,47} Diaz et al. proposed that AF-SkX has topological magnon bands and hosts topologically protected chiral magnonic edge states.⁴⁸ Fang et al. studied the possible magnetic parameters for the creation of AF-SkX.⁴⁹ Despite these theoretical models, at present, few 2D AFM materials have been demonstrated to have skyrmion physics.^{50–52} Actually, up to now, AF-SkX has not yet been realized in real 2D materials.

In this work, we show that an AF-SkX can be realized in a 2D lattice of Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ based on first-principles calculations and Monte Carlo (MC) simulations. Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ is demonstrated to exhibit a large Dzyaloshinskii–Moriya interaction (DMI) due to the joint effect of the broken inversion symmetry and strong spin–orbit coupling (SOC). This, along with the geometric frustration of its triangular lattice, stabilizes the skyrmion physics and AF-SkX under an external magnetic field. With the introduction of the skyrmion number Q and total chirality χ_L as measurements, the evaluation of AFM topological spin textures in Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ with the magnetic field and temperature is mapped out. Distinct from those reported in 3D systems, it can manifest two different antiferromagnetic skyrmion phases. In addition, through contact with Sc_2CO_2 , the AF-SkX in Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ can be created or annihilated by ferroelectric switching. Our works promote further study on AFM skyrmion physics in a 2D lattice.

RESULTS AND DISCUSSION

Figure 1a presents the crystal structure of Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, which is derived from the prototype MA_2Z_4 (M = early transition metal, A = Si, Ge, and Z = N, P, or As).^{53–57} Each unit cell contains one Cr, two Si, two N, and two As atoms, which are stacked in the sequence of N–Si–N–Cr–As–Si–As. It has a triangular lattice structure with the space group $P3m1$. Therefore, the inversion symmetry is broken. The optimized lattice constant of Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ is optimized to be 3.12 Å. The elastic constants of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ are found to be $C_{11} = C_{22} = 31.6 \text{ N m}^{-1}$, $C_{12} = C_{21} = -26.9 \text{ N m}^{-1}$, and $C_{66} = 31.2 \text{ N m}^{-1}$, which satisfy the Born–Huang criteria ($C_{11}C_{22} - C_{12}^2 > 0$ and $C_{66} > 0$),^{58–60} suggesting its mechanical stability. We also calculate the formation energy of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, which is defined as $E_f = E_{\text{tot}} - E_{\text{Cr}} - 2E_{\text{Si}} - 2E_{\text{N}} - 2E_{\text{As}}$. Here, E_{Cr} , E_{Si} , E_{N} , and E_{As} are the chemical potentials of Cr, Si, N, and As, respectively (taken from Cr bulk, silicene, N_2 , and arsenene). E_f is calculated to be -11.23 eV , indicating that it is thermodynamically favorable.⁶¹ In experiments, $\text{CrSi}_2\text{N}_2\text{As}_2$ is expected to be

synthesized by a chemical vapor deposition method, similar to its prototype MoSi_2N_4 .⁵³

The valence electronic configuration of Cr atom is $3d^5 4s^1$. By coordinating with three N and three As atoms in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, each Cr donates three electrons for bonding, resulting in the valence electronic configuration $3d^3 4s^0$. As shown in Figure 1a, the six neighboring N/As atoms form a distorted-trigonal-prismatic geometry for the Cr atom. In such a coordination environment, the d orbitals of the Cr atom roughly split into three groups, i.e., d_{z^2} , e_1 (d_{xy} , $d_{x^2-y^2}$), and e_2 (d_{xz} , d_{yz}). The three remaining valence electrons of Cr atom half-fill the d_{z^2} and e_1 (d_{xy} , $d_{x^2-y^2}$) orbitals, which would give rise to a magnetic moment of $3 \mu_B$. Our first-principles calculations confirm that the magnetic moment distributed on Cr atom is $2.92 \mu_B$. Therefore, monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ is a 2D magnetic material.

The existence of magnetic moments cannot guarantee spin ordering, and thus it is necessary to study the preferred spin configuration. Two magnetic couplings are considered: FM coupling and AFM coupling. Based on a 1×2 supercell, the AFM coupling spin configuration is 72.1 meV (i.e., 36.1 meV/fu) in lower energy than the FM coupling configuration, which indicates that the AFM spin configuration is more favorable. The energy difference is similar to those other reported systems, such as VS_2/VSe_2 ⁶² and CoGa_2S_4 .⁶³ The band structure and projected density of states (PDOS) of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ are shown in Figure S3. Obviously, it exhibits a metallic nature. The band structure around the Fermi level is contributed by Cr d and As p orbitals.

To further investigate the magnetic properties, we construct the Heisenberg type spin Hamiltonian model of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, which consists of the nearest sites isotropic Heisenberg exchange interaction, the anisotropic Heisenberg exchange interaction, DMI, the single-ion anisotropy (SIA), and the Zeeman interaction (see Note 2 in the Supporting Information for a detailed discussion or the model). The effective Hamiltonian can be expressed as

$$H = -J \sum_{\langle i,j \rangle} (\mathbf{S}_i \cdot \mathbf{S}_j) - \sum_{\langle i,j \rangle} d_{xy} (\mathbf{S}_i \times \mathbf{S}_j) - \lambda \sum_{\langle i,j \rangle} (S_i^z S_j^z) - K \sum_i (S_i^z)^2 - h \sum_i S_i^z$$

Here, $\mathbf{S}_{i/j}$ is the unit vector ($|\mathbf{S}| = 1$) corresponding to local spin at the $i_{\text{th}}/j_{\text{th}}$ Cr atom. d_{xy} and K indicate the magnetic parameters characterizing in-plane DMI and SIA, respectively. The Heisenberg exchange interaction in the model is divided into two parts: i.e., isotropic and anisotropic terms. J is the isotropic exchange interaction parameter ($J > 0$ indicates

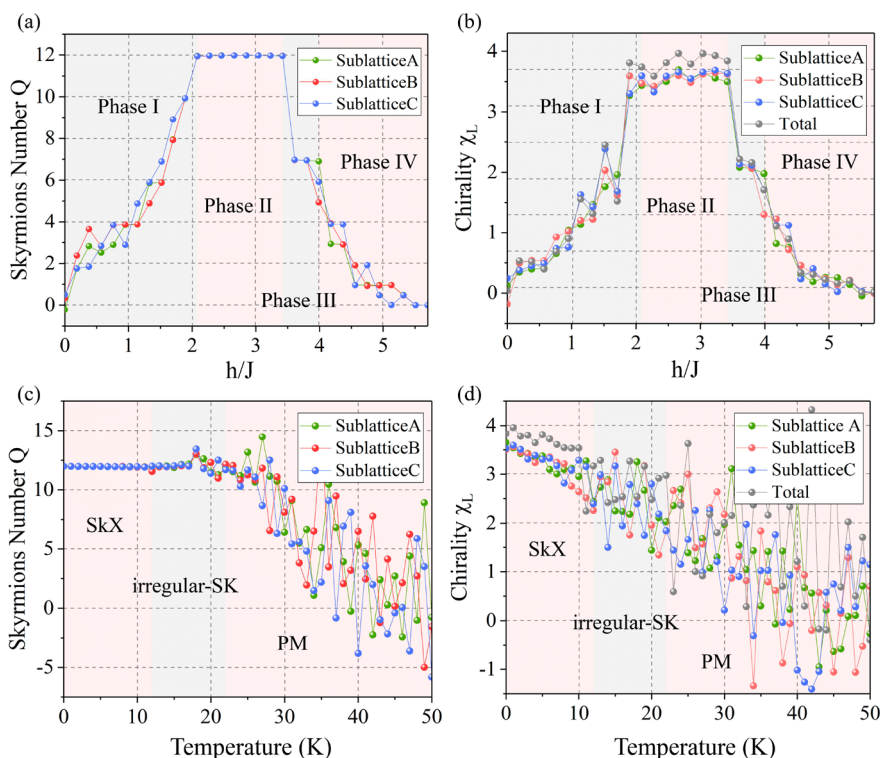


Figure 2. Skyrmion number Q (a) and chirality χ_L (b) as a function of external magnetic field for monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. Skyrmion number Q (c) and chirality χ_L (d) as a function of temperature for monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ monolayer under $h/J = 2.85$.

ferromagnetic coupling), and λ is the anisotropic parameter. In the last term, h indicates the strength of the external magnetic field along z axis.

The isotropic exchange interaction parameter J is calculated to be -9.013 meV, which indicates an AFM neighboring interaction for monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. It should be noted that the magnitude of J is comparable to those of NiI_2 ,⁶⁴ CrX ($X = \text{P, As, and Sb}$),⁶⁵ and FeOOH .⁶⁶ As schematically shown in Figure 1b, such an AFM neighboring interaction in a triangular lattice cannot result in an antiparallel arrangement for the magnetic moments of every two atoms at vertices of a unit equilateral triangle. This would lead to geometric frustration in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. Namely, the Cr lattice presents a three-sublattice 120° structure,⁶⁷ wherein the spins in each sublattice (colored in Figure 1c) are parallel arranged in parallel.

According to Moriya's rule,⁶⁸ the symmetry of triangular lattice enforces that D_{ij} for the nearest-neighboring Cr atoms is perpendicular to their bond. The in-plane DMI parameter d_{xy} is calculated to be 1.960 meV (detailed in Note 4 in the Supporting Information). Such a large value can be attributed to the combined effect of the large SOC strength within the As atom and the large electronegativity difference between N and As atoms. This results in $d_{xy}/J = 0.217$ for monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, which is located in the typical range for the formation of skyrmions (around 0.1 – 0.2).⁶⁹ The SIA parameter K is calculated to be -0.459 meV, indicating the in-plane magnetic anisotropy. The anisotropic exchange interaction parameter λ is estimated to be 0.122 , which favors out-of-plane magnetic anisotropy. Obviously, K plays a dominant role, estimating the in-plane magnetic anisotropy for monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. In this regard, monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ holds the possibility for hosting nontrivial topological spin structures.

Based on the first-principles calculations parametrized Hamiltonian, we perform parallel tempering MC simulations to explore the spin configurations in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. As displayed in Figure S5, it exhibits a striped-like spin texture (referred to as the spin spiral (SS) phase). We then investigate its spin structures under different external magnetic fields and temperatures. Here, we introduce the skyrmion number Q and chirality χ_L to detect the spin textures effectively. The skyrmion number Q and chirality χ_L are defined as³²

$$Q = \frac{1}{4\pi} \sum_n q$$

$$\chi_L = \frac{1}{8\pi} \sum_n [\chi_{L,r_n^{(1)}} + \chi_{L,r_n^{(2)}}]$$

with

$$\tan \frac{q}{2} = A_{L,r_n^{(1)}} \chi_{L,r_n^{(1)}} + A_{L,r_n^{(2)}} \chi_{L,r_n^{(2)}}$$

Here,

$A_{L,r_n^{(1,2)}} = 1/[1 + \mathbf{s}_i^{(1,2)} \cdot \mathbf{s}_j^{(1,2)} + \mathbf{s}_j^{(1,2)} \cdot \mathbf{s}_k^{(1,2)} + \mathbf{s}_k^{(1,2)} \cdot \mathbf{s}_i^{(1,2)}]$ is calculated in the n th unit equilateral triangle of the lattice, and the upper corner marks 1 and 2 represent two corner-shared triangles (highlighted in Figure 1c). $\chi_{L,r_n^{(1,2)}} = \mathbf{s}_i^{(1,2)} \cdot (\mathbf{s}_j^{(1,2)} \times \mathbf{s}_k^{(1,2)})$ is the chirality localized in the unit equilateral triangle. $\mathbf{s}_i^{(1,2)}$, $\mathbf{s}_j^{(1,2)}$, and $\mathbf{s}_k^{(1,2)}$ are the three spin vectors in the anticlockwise lattice. We calculate the Q and χ_L values in three sublattices of the triangular lattice, wherein the spin vectors run in each sublattice. Also, we calculate the total local chirality without the sublattice distinction.

Figure 2 presents the skyrmion number Q and chirality χ_L of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ as a function of the external magnetic

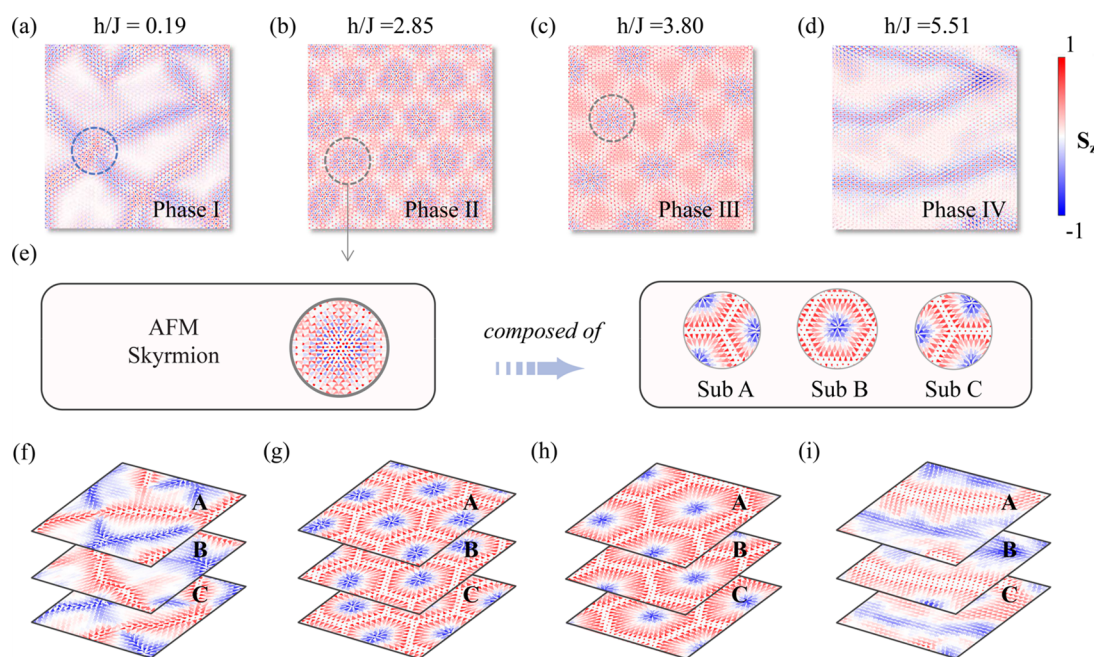


Figure 3. Specific spin textures in the triangular lattice and sublattices of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ under (a, f) phase I, (b, g) phase II, (c, h) phase III, and (d, i) phase IV. (e) Spin patterns of an isolated AFM skyrmion (left panel) and three FM skyrmions in sublattices A, B, and C (right panel).

field. It can be seen that both Q and χ_L increase gradually under an increasing external magnetic field h/J from 0 to 2.08. Such a continuous change suggests no phase transition, and thus phase I is the SS phase. By further increasing h/J to 3.42, intriguingly, Q of the three sublattices share the same values and remain unchanged at a platform, and χ_L values are also stable with a slight fluctuation. These characters indicate that the topologically protected AF-SkX phase probably occurs. Upon increasing h/J from 3.42 to 3.99, Q values of the three sublattices jump to another platform and remain unchanged, and χ_L values remain stable with a slight fluctuation as well, suggesting another possible AF-SkX phase. Therefore, monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ might exhibit two AF-SkX phases. On further increasing h/J to more than 3.99, Q values of the three sublattices are becoming different and gradually reduce to zero. This process corresponds to the AF-SkX phase becoming broken and being evaluated finally as the SS phase.

To get the specific spin patterns of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ under an external magnetic field, taking $h/J = 0.19$, 2.85, 3.80, and 5.51 as examples of these four phases, we show their spin structures in Figure 3. For more examples, please refer to Figure S5. Under $h/J = 0.19$, the spin structure of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ is roughly similar to the case under $h/J = 0$, displaying an AFM SS phase (see Figure 3a). As shown in Figure 3f, such an AFM SS phase can be considered to be the interpenetration of the FM SS configurations in the three sublattices. From the spin texture shown in Figure 3a, no skyrmion can be observed; however, Q is calculated to be nonzero. Such a nonzero Q is correlated to the vortices formed at the intersections of different stripes as Q is characterized by the product of vortex and polarity. Meanwhile, as shown in Figure S5, with rising h/J in phase I, the number of intersections increases and thus Q increases.

For the case under $h/J = 2.85$, as displayed in Figure 3b, a regular array of AFM skyrmions is observed, giving rise to the intriguing AF-SkX phase. The AF-SkX is interpenetrated by the

FM skyrmion crystals of its three sublattices (see Figure 3g). For these three FM skyrmion crystals, the skyrmions are present in the same size and density, and they can be converted to each other under a translation operation. From Figure 3e, it can be seen that each AFM skyrmion in AF-SkX is formed by a FM skyrmion from one sublattice and two striped-like textures from the other two sublattices. Therefore, phase II is the topologically protected AF-SkX phase. Figure 3c,h presents the spin structure of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ under $h/J = 3.80$. Obviously, it exhibits a similar topologically protected spin texture: namely, the AF-SkX phase. As compared with the case under $h/J = 2.85$, Q of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ under $h/J = 3.80$ is smaller, and thus its skyrmions are present in a smaller density. Accordingly, monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ hosts two topologically protected AF-SkX phases. By increasing h/J to greater than 3.80, as shown in Figure S5, the array of AFM skyrmions in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ begins to be irregular, leading to the broken AF-SkX phase. Subsequently, the AFM skyrmions disappear and sparse stripes appear, which yields the SS phase (Figure 3d,i). Such an existence of AF-SkX in Janus monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ is consistent with the previous theories studying emergent exotic phases in the antiferromagnetic triangular lattice.^{32,46,47}

We then take the case under $h/J = 2.85$ as an example to investigate the influence of temperature on the nontrivial topological spin texture in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$. As shown in Figure 2c, with an increase in the temperature from 0 to 12 K, Q values are almost unchanged, and the differences in Q among the three sublattices are negligible. However, for χ_L values, although they have a noticeable decrease, the differences among the three sublattices are tiny (see Figure 2d). These features indicate that the AF-SkX phase in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ can be preserved with the temperature reaching up to ~ 12 K. As the temperature is increased to greater than 12 K, both Q and χ_L begin to decrease with oscillation (Figure 2c,d), which corresponds to the array of

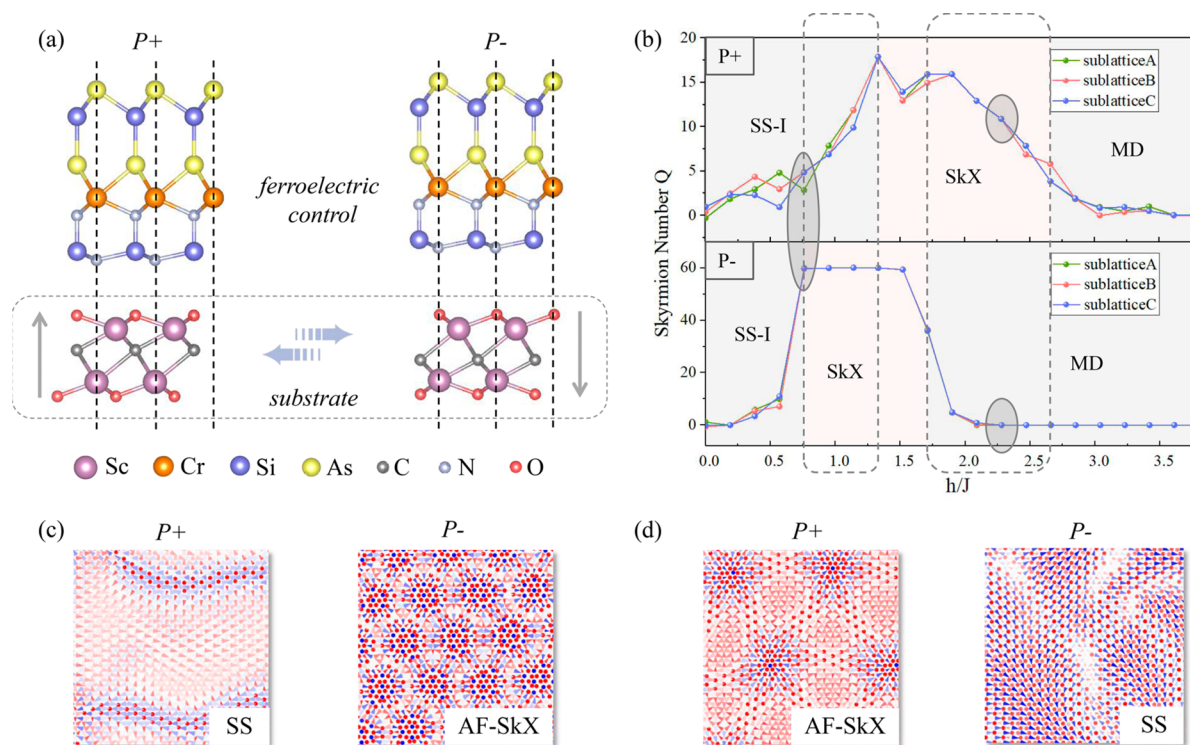


Figure 4. (a) Crystal structures of heterobilayer $\text{CrSi}_2\text{N}_2\text{As}_2/\text{Sc}_2\text{CO}_2$ under $P+$ (left panel) and $P-$ (right panel). (b) Skyrmion number Q as a function of h/J for $P+$ and $P-$. Specific spin patterns for $P+$ and $P-$ with (c) $h/J = 0.76$ and (d) $h/J = 2.27$.

AFM skyrmions being irregular (called an irregular skyrmion (irregular-SK) phase). When the temperature is greater than ~ 22 K, the oscillations are chaotic and the amplitudes are huge, suggesting that the regular AF-SkX in monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$ is annihilated thoroughly and the paramagnetic (PM) phase appears.

Considering that the ferroelectric substrate can affect the magnetic parameters,^{20,70} we next propose a mechanism of coupling topological spin structures with ferroelectricity for realizing electrical control of AF-SkX. Here, monolayer Sc_2CO_2 is selected as the ferroelectric substrate. Given the switchable polarization of Sc_2CO_2 , there are two types of heterobilayer $\text{CrSi}_2\text{N}_2\text{As}_2/\text{Sc}_2\text{CO}_2$: one with the electric polarization pointing along the $+z$ direction ($P+$) and the other with electric polarization pointing along the $-z$ direction ($P-$). Figure 4a shows the crystal structures of heterobilayer $\text{CrSi}_2\text{N}_2\text{As}_2/\text{Sc}_2\text{CO}_2$. We mainly discuss the N-side interface case, and the As-side interface case is shown in the Supporting Information.

The corresponding magnetic parameters of $P+$ and $P-$ are summarized in Table 2. It can be seen that in compared with the freestanding case, the magnetic moment distributed on the Cr atom is slightly increased to $3.03 \mu_B$ for both $P+$ and $P-$. It should be noted that the exchange interaction parameters J of $P+$ (-3.061 meV) and $P-$ (-5.744 meV) are reduced as compared with that of monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, which can be attributed to the proximity effect (see Note 12 in the Supporting Information). This implies that AF-SkX would exist at a relatively lower temperature. To stabilize AF-SkX in $P+$ and $P-$ under the relatively high temperature, a promising strategy is to apply a moderate magnetic field.^{3,71,72} Similarly to the exchange interaction, the value of d_{xy} is reduced slightly for $P+$ (1.355 meV) and $P-$ (1.439 meV). This indicates that the DMIs for both $P+$ and $P-$ are weakened under the proximity

effect of Sc_2CO_2 . The anisotropic Heisenberg exchange interaction, as given in Tables 1 and 2, is also weakened

Table 1. Isotropic Heisenberg Exchange Interaction Parameter J , in-Plane DMI Parameter d_{xy} , Anisotropic Exchange Interaction Parameter λ , SIA Parameter K , and Magnetic Moment m on Cr Atom for Monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$

	J (meV)	d_{xy} (meV)	λ (meV)	K (meV)	m (μ_B)
$\text{CrSi}_2\text{N}_2\text{As}_2$	-9.013	1.960	0.122	-0.459	2.92

Table 2. Lattice Constant a , Isotropic Heisenberg Exchange Interaction Parameter J , in-Plane DMI Parameter d_{xy} , Anisotropic Exchange Interaction Parameter λ , SIA Parameter K , and Magnetic Moment m on Cr Atom for $\text{CrSi}_2\text{N}_2\text{As}_2/\text{Sc}_2\text{CO}_2$ Heterobilayer under $P+$ and $P-$

	a (Å)	J (meV)	d_{xy} (meV)	λ (meV)	K (meV)	m (μ_B)
$P+$	3.29	-3.061	1.355	0.080	-0.475	3.03
$P-$	3.29	-5.744	1.439	0.100	-0.450	3.03

under the proximity effect. However, for the SIA energy, the change is slight for both $P+$ (-0.475 meV) and $P-$ (-0.450 meV). Figure S4 displays the band structures and PDOSs of $P+$ and $P-$. Also, as compared with monolayer $\text{CrSi}_2\text{N}_2\text{As}_2$, the electronic properties are significantly modified due to the proximity effect. As the magnetic parameters of $P+$ and $P-$ are different, $P+$ and $P-$ would have different topological spin structures, implying the possible ferroelectric control of spin textures in heterobilayer $\text{CrSi}_2\text{N}_2\text{As}_2/\text{Sc}_2\text{CO}_2$.

Figure 4b shows the skyrmion number Q of $P+$ and $P-$ as a function of h/J (To compare the variation of spin configurations under the same external magnetic fields, the parameter J in h/J is the value of monolayer in all cases of a

heterobilayer). For P+, the skyrmion number Q in the three sublattices rises gradually upon increasing the external magnetic field h/J from 0 to 1.33. The spin structures of P+ under $h/J = 0$ –1.33 are shown in Figure S6, which display an AFM SS phase. Under $h/J = 1.33$ –2.66, Q has a large value. From the corresponding spin structures shown in Figure S6, we can see that P+ displays a AF-SkX phase. This indicates that a transition from the SS phase to the protected topologically AF-SkX phase occurs under $h/J = 1.33$ for P+. When h/J is increased to greater than 2.66, Q is gradually reduced to zero, which corresponds to the AF-SkX phase transferring to SS phase for P+. Such a variation of spin textures under h/J is shared by the case of P–, but the corresponding critical points are different, i.e., $h/J = 0.76$ and 1.70, as shown in Figure 4b and Figure S7.

Obviously, under $h/J = 0.76$ –1.33 and 1.70–2.66, the creation and annihilation of the AF-SkX phase can be achieved by ferroelectric polarization switching. Here, taking $h/J = 0.76$ and 1.70 as examples of these two regions, we display their spin structures in Figure 4c,d. It can be seen that, under $h/J = 0.76$, SS and AF-SkX spin patterns appear in P+ and P–, respectively. Under the ferroelectric transition, P+ can be transformed into P–, which is accompanied by the transition of an SS spin pattern into an AF-SkX spin pattern. However, for the case under $h/J = 1.70$, an AF-SkX spin pattern appears in P+, which can be readily reversed to the SS phase upon switching the ferroelectric polarization. In this regard, the ferroelectrically tunable AF-SkX is realized in monolayer CrSi₂N₂As₂.

For comparison, we illustrate the magnetic parameters and specific spin patterns of the As-side interface case in Note 9 of the Supporting Information. From Table 2 and Table S1, we find that the magnetic parameters of the As-side heterobilayer are similar to the case for the N-side, only with slight differences in exchange coupling and DMI parameters. This can be attributed to the long distance between Cr atoms and the substrate. By comparing the spin patterns shown in Figures S6–S8, it can be seen that the spin patterns of the As-side and N-side cases share roughly similar trends for the response to an external magnetic field, but only one AF-SkX phase emerges in the As-side case. This results from the relatively small antiferromagnetic coupling in As-side case.

Finally, we discuss the strain effect on the magnetic parameters as well as the topological spin structures of monolayer CrSi₂N₂As₂. As shown in Figure S11, with increasing strain, J increases monotonically, while d_{xy} decreases monotonically, leading to the monotonic decrease of d_{xy}/J . This indicates that the topological spin structures of monolayer CrSi₂N₂As₂ would also vary with strain. Figure S12 shows the specific spin patterns of monolayer CrSi₂N₂As₂ under different strains and h/J values. We can see that the AF-SkX phase appears under $h/J = 0.76$, 1.33, and 2.66 for the cases of 3%, 1%, and –1% strain, respectively. However for the case of –3% strain, the AF-SkX phase is not observed within the considered h/J . By comparing with the pure case, it is interesting to note that strain can be used to significantly reduce the required h/J to realize the AF-SkX phase. Moreover, as shown in Figure S12, along with increasing strain, the size of AFM skyrmions in AF-SkX is reduced, which is favorable for practical applications.

CONCLUSIONS

In summary, using first-principles calculations and MC simulations, we demonstrate the existence of the AF-SkX phase in the 2D lattice of monolayer CrSi₂N₂As₂. The skyrmion physics in monolayer CrSi₂N₂As₂ is correlated to its large DMI and the geometric frustration of its triangular lattice. The evaluation of topological spin textures in monolayer CrSi₂N₂As₂ with h/J and temperature is comprehensively investigated. Distinct from those reported in 3D systems, two different antiferromagnetic skyrmion phases can be manifested in monolayer CrSi₂N₂As₂. Furthermore, with interfacing to monolayer Sc₂CO₂, the creation and annihilation of AF-SkX in monolayer CrSi₂N₂As₂ are achieved by ferroelectric polarization switching. The explored phenomena and insights greatly enrich the AFM skyrmion physics in a 2D lattice.

METHODS

DFT Method. Our first-principles calculations are performed based on density functional theory (DFT) as implemented in the Vienna ab initio simulation package (VASP).^{73,74} The projector augmented wave (PAW) method is adopted to treat the ionic potential.⁷⁵ The Perdew–Burke–Ernzerhof (PBE) functional of the generalized gradient approximation (GGA) is used for the exchange–correlation interactions.⁷⁶ To describe the strong correlations of 3d electrons well, the GGA+ U method is adopted with an effective Hubbard $U = 3$ eV for the 3d electrons of a Cr atom.⁷⁷ Taking $U = 5.0$ eV as an example, we perform test calculations to check the effect of Hubbard U value and find that the results are roughly similar to the main conclusions based on $U = 3.0$ eV (see Note 13 in the Supporting Information). The plane-wave cutoff energy is set to 520 eV. The convergence criteria for force and energy are set to 0.001 eV/Å and 10^{–6} eV, respectively. The Monkhorst–Pack k -point mesh of $13 \times 13 \times 1$ is adopted to sample the 2D Brillouin zone. The DFT-D3 method is employed for treating van der Waals interactions in the heterostructures. Phonon spectra are obtained by the PHONOPY code based on a $4 \times 4 \times 1$ supercell.⁷⁸

MC Simulations. Using the magnetic parameters obtained from first-principles calculations, parallel tempering MC simulations with the Metropolis algorithm are carried out to get the energy minimum spin textures.⁷⁹ The spin textures are obtained based on a $120 \times 120 \times 1$ supercell with 1×10^5 MC steps performed for each temperature (from 660 K gradually decreasing in steps of 5 K to the investigated low temperature).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.2c08544>.

Phonon spectra and ELF, discussion of the model, validation of the theoretical method, calculation details of magnetic parameters, skyrmion number Q , electronic properties, detailed spin patterns of monolayer CrSi₂N₂As₂, detailed spin patterns of the heterostructure CrSi₂N₂As₂/Sc₂CO₂, spin textures of As-side cases, specific spin patterns and total magnetic moments at different temperatures, effect of strain, effect of electric field, effect of Hubbard U values, and magnetic properties of CrSi₂N₂P₂ and CrSi₂P₂As₂ (PDF)

AUTHOR INFORMATION

Corresponding Authors

Ying Dai – School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, People's

Republic of China; orcid.org/0000-0002-8587-6874;
Email: daiy60@sdu.edu.cn

Yandong Ma – School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, People's Republic of China; orcid.org/0000-0003-1572-7766; Email: yandong.ma@sdu.edu.cn

Authors

Kaiying Dou – School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, People's Republic of China

Wenhui Du – School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, People's Republic of China

Zhonglin He – School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, People's Republic of China

Baibiao Huang – School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, People's Republic of China; orcid.org/0000-0002-0416-944X

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsnano.2c08544>

Author Contributions

K.D. executed calculations. K.D., W.D., Z.H., and Y.D. performed data analysis. K.D. and Y.M. cowrote the paper. Y.D., B.H., and Y.M. supervised the project. All authors commented on the paper.

Notes

The preprint version of this work is available at <https://arxiv.org/abs/2204.08792>.⁸⁰

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Nos. 12274261 and 12074217), the Shandong Provincial Science Foundation for Excellent Young Scholars (No. ZR2020YQ04), the Shandong Provincial Natural Science Foundation (Nos. ZR2019QA011), the Shandong Provincial Key Research and Development Program (Major Scientific and Technological Innovation Project) (No. 2019JZZY010302), the Shandong Provincial QingChuang Technology Support Plan (No. 2021KJ002), and the Qilu Young Scholar Program of Shandong University.

REFERENCES

- (1) Yu, X.; Onose, Y.; Kanazawa, N.; Park, J.; Han, J.; Matsui, Y.; Nagaosa, N.; Tokura, Y. Real-space observation of a two-dimensional skyrmion crystal. *Nature* **2010**, *465*, 901–904.
- (2) Duine, R. Skyrmions singled out. *Nat. Nanotechnol.* **2013**, *8*, 800–802.
- (3) Tokura, Y.; Kanazawa, N. Magnetic skyrmion materials. *Chem. Rev.* **2021**, *121*, 2857–2897.
- (4) Jiang, W.; Chen, G.; Liu, K.; Zang, J.; Velthuis, S.; Hoffmann, A. Skyrmions in magnetic multilayers. *Phys. Rep.* **2017**, *704*, 1–49.
- (5) Sampaio, J.; Cros, V.; Rohart, S.; Thiaville, A.; Fert, A. Nucleation, stability and current-induced motion of isolated magnetic skyrmions in nanostructures. *Nat. Nanotechnol.* **2013**, *8*, 839–844.
- (6) Parkin, S. S. P.; Hayashi, M.; Thomas, L. Magnetic domain-wall racetrack memory. *Science* **2008**, *320*, 190–194.
- (7) Müller, J. Magnetic skyrmions on a two-lane racetrack. *New J. Phys.* **2017**, *19*, No. 025002.

(8) Kang, W.; Huang, Y.; Zheng, C.; Lv, W.; Lei, N.; Zhang, Y.; Zhang, X.; Zhou, Y.; Zhao, W. Voltage controlled magnetic skyrmion motion for racetrack memory. *Sci. Rep.* **2016**, *6*, 23164.

(9) Fert, A.; Cros, V.; Sampaio, J. Skyrmions on the track. *Nat. Nanotechnol.* **2013**, *8*, 152–156.

(10) Yang, H.; Thiaville, A.; Rohart, S.; Fert, A.; Chshiev, M. Anatomy of Dzyaloshinskii-Moriya interaction at Co/Pt interfaces. *Phys. Rev. Lett.* **2015**, *115*, 267210.

(11) Moreau-Luchaire, C.; Moutafis, C.; Reyren, N.; Sampaio, J.; Vaz, C.; Horne, N.; Bouzehouane, K.; Garcia, K.; Deranlot, C.; Warnicke, P.; Wohlhuter, P.; George, J.; Weigand, M.; Raabe, J.; Cros, V.; Fert, A. Additive interfacial chiral interaction in multilayers for stabilization of small individual skyrmions at room temperature. *Nat. Nanotechnol.* **2016**, *11*, 444–448.

(12) Wang, L.; Feng, Q.; Kim, Y.; Kim, R.; Lee, K.; Pollard, S.; Shin, Y.; Zhou, H.; Peng, W.; Lee, D.; Meng, W.; Yang, H.; Han, J.; Kim, M.; Lu, Q.; Noh, T. Ferroelectrically tunable magnetic skyrmions in ultrathin oxide heterostructures. *Nat. Mater.* **2018**, *17*, 1087–1094.

(13) Romming, N.; Kubetzka, A.; Hanneken, C.; Bergmann, K.; Wiesendanger, R. Field-dependent size and shape of single magnetic skyrmions. *Phys. Rev. Lett.* **2015**, *114*, 177203.

(14) Deng, Y.; Yu, Y.; Song, Y.; Zhang, J.; Wang, N.; Sun, Z.; Yi, Y.; Wu, Y.; Wu, S.; Zhu, J.; Wang, J.; Chen, X.; Zhang, Y. Gate-tunable room-temperature ferromagnetism in two-dimensional Fe₃GeTe₂. *Nature* **2018**, *563*, 94–99.

(15) Yu, X.; Kanazawa, Z.; Onose, Y.; Kimoto, K.; Zhang, W.; Ishiwata, S.; Matsui, Y.; Tokura, Y. Near room-temperature formation of a skyrmion crystal in thin-films of the helimagnet FeGe. *Nat. Mater.* **2011**, *10*, 106–109.

(16) Boule, O.; Vogel, J.; Yang, H.; Pizzini, S.; Chaves, D.; Locatelli, A.; Menteş, T.; Sala, A.; Buda-Prejbeanu, L.; Klein, O.; Belmeguenai, M.; Roussigné, Y.; Stashkevich, A.; Chérif, S.; Aballe, L.; Foerster, M.; Chshiev, M.; Auffret, S.; Miron, I.; Gaudin, G. Room-temperature chiral magnetic skyrmions in ultrathin magnetic nanostructures. *Nat. Nanotechnol.* **2016**, *11*, 449–454.

(17) Soumyanarayanan, M.; Gonzalez Oyarce, A.; Tan, A.; Im, M.; Petrovic, A.; Ho, P.; Khoo, K.; Tran, M.; Gan, C.; Ernult, F.; Panagopoulos, C. Tunable room-temperature magnetic skyrmions in Ir/Fe/Co/Pt multilayers. *Nat. Mater.* **2017**, *16*, 898–904.

(18) Xu, C.; Chen, P.; Tan, H.; Yang, Y.; Xiang, H.; Bellaiche, L. Electric-field switching of magnetic topological charge in type-I multiferroics. *Phys. Rev. Lett.* **2020**, *125*, No. 037203.

(19) Sun, W.; Wang, W.; Li, H.; Zhang, G.; Chen, D.; Wang, J.; Cheng, Z. Controlling bimerons as skyrmion analogues by ferroelectric polarization in 2D van der Waals multiferroic heterostructures. *Nat. Commun.* **2020**, *11*, 5930.

(20) Huang, K.; Shao, D.; Tsybmal, E. Ferroelectric control of magnetic skyrmions in two-dimensional van der Waals heterostructures. *Nano Lett.* **2022**, *22*, 3349–3355.

(21) Liang, X.; Xia, J.; Zhang, X.; Ezawa, M.; Tretiakov, O.; Liu, X.; Qiu, L.; Zhao, G.; Zhou, Y. Antiferromagnetic skyrmion-based logic gates controlled by electric currents and fields. *Appl. Phys. Lett.* **2021**, *119*, No. 062403.

(22) Zhang, X.; Zhou, Y.; Ezawa, M. Antiferromagnetic skyrmion: stability, creation and manipulation. *Sci. Rep.* **2016**, *6*, 24795.

(23) Barker, J.; Tretiakov, O. Static and dynamical properties of antiferromagnetic skyrmions in the presence of applied current and temperature. *Phys. Rev. Lett.* **2016**, *116*, 147203.

(24) Jungwirth, T.; Marti, X.; Wadley, P.; Wunderlich, J. Antiferromagnetic spintronics. *Nat. Nanotechnol.* **2016**, *11*, 231–241.

(25) Jungfleisch, M.; Zhang, W.; Hoffmann, A. Perspectives of antiferromagnetic spintronics. *Phys. Lett. A* **2018**, *382*, 865–871.

(26) Baltz, V.; Manchon, A.; Tsoi, M.; Moriyama, T.; Ono, T.; Tserkovnyak, Y. Antiferromagnetic spintronics. *Rev. Mod. Phys.* **2018**, *90*, No. 015005.

(27) Cheng, R.; Xiao, J.; Niu, Q.; Brataas, A. Spin pumping and spin-transfer torques in antiferromagnets. *Phys. Rev. Lett.* **2014**, *113*, No. 057601.

- (28) Železný, J.; Wadley, P.; Olejník, K.; Hoffmann, A.; Ohno, H. Spin transport and spin torque in antiferromagnetic devices. *Nat. Phys.* **2018**, *14*, 220–228.
- (29) Liu, Z.; Dias, M.; Lounis, S. Theoretical investigation of antiferromagnetic skyrmions in a triangular monolayer. *J. Phys.: Condens. Matter* **2020**, *32*, 425801.
- (30) Osorio, S.; Rosales, H.; Sturla, M.; Cabra, D. Composite spin crystal phase in antiferromagnetic chiral magnets. *Phys. Rev. B* **2017**, *96*, No. 024404.
- (31) Li, X.; Shen, L.; Bai, Y.; Wang, J.; Zhang, X.; Xia, J.; Ezawa, M.; Tretiakov, O.; Xu, X.; Mruczkiewicz, M.; Krawczyk, M.; Xu, Y.; Chantrell, R.; Zhou, Y. Bimeron clusters in chiral antiferromagnets. *npj Comput. Mater.* **2020**, *6*, 169.
- (32) Rosales, H.; Cabra, D.; Pujol, P. Three-sublattice skyrmion crystal in the antiferromagnetic triangular lattice. *Phys. Rev. B* **2015**, *92*, 214439.
- (33) Keesman, R.; Raaijmakers, M.; Baerends, A.; Barkema, G.; Duine, R. Skyrmions in square-lattice antiferromagnets. *Phys. Rev. B* **2016**, *94*, No. 054402.
- (34) Villalba, M.; Albarracín, F.; Rosales, H.; Cabra, D. Field-induced pseudo-skyrmion phase in the antiferromagnetic kagome lattice. *Phys. Rev. B* **2019**, *100*, 245106.
- (35) Mukherjee, A.; Kathyat, D.; Kumar, S. Antiferromagnetic skyrmions and skyrmion density wave in a Rashba-coupled Hund insulator. *Phys. Rev. B* **2021**, *103*, 134424.
- (36) Mukherjee, A.; Kathyat, D.; Kumar, K. Engineering antiferromagnetic skyrmions and antiskyrmions at metallic interfaces. *Phys. Rev. B* **2022**, *105*, No. 075102.
- (37) Jani, H.; Lin, J.; Chen, J.; Harrison, J.; Maccherozzi, F.; Schäd, J.; Prakash, S.; Eom, C.; Ariando, A.; Venkatesan, T.; Radaelli, P. Antiferromagnetic half-skyrmions and bimerons at room temperature. *Nature* **2021**, *590*, 74–79.
- (38) Gao, S.; Rosales, H.; Albarracín, F.; Tsurkan, V.; Kaur, G.; Fennell, T.; Steffens, P.; Boehm, M.; Cermak, P.; Schneidewind, A.; Ressouche, E.; Cabra, D.; Ruegg, C.; Zaharko, O. Fractional antiferromagnetic skyrmion lattice induced by anisotropic couplings. *Nature* **2020**, *586*, 37–41.
- (39) Li, Y.; Feng, Q.; Li, S.; Huang, K.; Ma, M.; Gan, W.; Zhou, H.; Jin, X.; Wang, X.; Lu, Y. An artificial skyrmion platform with robust tunability in synthetic antiferromagnetic multilayers. *Adv. Funct. Mater.* **2020**, *30*, 1907140.
- (40) Chen, R.; Cui, Q.; Hann, L.; Xue, X.; Liang, J.; Bai, H.; Zhou, Y.; Pan, F.; Yang, H.; Song, C. Controllable generation of antiferromagnetic skyrmions in synthetic antiferromagnets with thermal effect. *Adv. Funct. Mater.* **2022**, *32*, 2111906.
- (41) Legrand, W.; Maccariello, D.; Ajejas, F.; Collin, S.; Vecchiola, A.; Bouzehouane, K.; Reyren, N.; Cros, V.; Fert, A. Room-temperature stabilization of antiferromagnetic skyrmions in synthetic antiferromagnets. *Nat. Mater.* **2020**, *19*, 34–42.
- (42) Kim, K.; Lim, S. Y.; Kim, J.; Lee, J.-U.; Lee, S.; Kim, P.; Park, K.; Son, S.; Park, C.-H.; Park, J.-G.; Cheong, H. Antiferromagnetic ordering in van der Waals 2D magnetic material MnPS₃ probed by Raman spectroscopy. *2D Mater.* **2019**, *6*, No. 041001.
- (43) Hu, L.; Wu, X.; Yang, J. Mn₂C monolayer: a 2D antiferromagnetic metal with high Néel temperature and large spin-orbit coupling. *Nanoscale* **2016**, *8*, 12939–12945.
- (44) Liu, J.; Meng, S.; Sun, J. Spin-orientation-dependent topological states in two-dimensional antiferromagnetic NiTi₂S₄ monolayers. *Nano Lett.* **2019**, *19*, 3321–3326.
- (45) Xu, X.; Ma, Y.; Zhang, T.; Lei, C.; Huang, B.; Dai, Y. Prediction of two-dimensional antiferromagnetic ferroelasticity in an AgF₂ monolayer. *Nanoscale Horiz.* **2020**, *5*, 1386–1393.
- (46) Mohylina, M.; Busa, J., Jr.; Zukovic, M. Formation and growth of skyrmion crystal phase in a frustrated Heisenberg antiferromagnet with Dzyaloshinskii-Moriya interaction. *J. Magn. Magn. Mater.* **2021**, *527*, 167755.
- (47) Mohylina, M.; Zukovic, M. Stability of skyrmion crystal phase in antiferromagnetic triangular lattice with DMI and single-ion anisotropy. *J. Magn. Magn. Mater.* **2022**, *546*, 168840.
- (48) Díaz, S.; Klinovaja, J.; Loss, D. Topological magnons and edge states in antiferromagnetic skyrmion crystals. *Phys. Rev. Lett.* **2019**, *122*, 187203.
- (49) Fang, W.; Raeliarijaona, A.; Chang, P.; Kovalev, A.; Belashchenko, K. Spirals and skyrmions in antiferromagnetic triangular lattices. *Phys. Rev. Mater.* **2021**, *5*, No. 054401.
- (50) Cui, Q.; Zhu, Y.; Ga, Y.; Liang, J.; Li, P.; Yu, D.; Cui, P.; Yang, H. Anisotropic Dzyaloshinskii-Moriya interaction and topological magnetism in two-dimensional magnets protected by P4m2 crystal symmetry. *Nano Lett.* **2022**, *22*, 2334–2341.
- (51) Mazurenko, V.; Rudenko, A.; Nikolaev, S.; Medvedeva, D.; Lichtenstein, A.; Katsnelson, M. Role of direct exchange and Dzyaloshinskii-Moriya interactions in magnetic properties of graphene derivatives C₂F and C₂H. *Phys. Rev. B* **2016**, *94*, 214411.
- (52) Ramadan, F.; Santos, F.; Drissi, L.; Lounis, S. Complex magnetism of the two-dimensional antiferromagnetic Ge₂F from a Neel spin-texture to a potential antiferromagnetic skyrmion. *RSC Adv.* **2021**, *11*, 8654.
- (53) Hong, Y.; Liu, Z.; Wang, L.; Zhou, T.; Ma, W.; Xu, C.; Feng, S.; Chen, L.; Chen, M.; Sun, D.; Chen, X.; Cheng, H.; Ren, W. Chemical vapor deposition of layered two-dimensional MoSi₂N₄ materials. *Science* **2020**, *369*, 670–674.
- (54) Wang, L.; Shi, Y.; Liu, M.; Zhang, A.; Hong, Y.; Li, R.; Gao, Q.; Chen, M.; Ren, W.; Cheng, H.; Li, Y.; Chen, X. Intercalated architecture of MA₂Z₄ family layered van der Waals materials with emerging topological, magnetic and superconducting properties. *Nat. Commun.* **2021**, *12*, 2361.
- (55) Zhong, H.; Xiong, W.; Lv, P.; Yu, J.; Yuan, S. Strain-induced semiconductor to metal transition in MA₂Z₄ bilayers (M = Ti, Cr, Mo; A = Si; Z = N, P). *Phys. Rev. B* **2021**, *103*, No. 085124.
- (56) Liu, Y.; Zhang, T.; Dou, K.; Du, W.; Peng, R.; Dai, Y.; Huang, B.; Ma, Y. Valley-contrasting physics in single-layer CrSi₂N₄ and CrSi₂P₄. *J. Phys. Chem. Lett.* **2021**, *12*, 8341–8346.
- (57) Rezavand, A.; Ghobadi, N.; Behnamghader, B. Electronic and spintronic properties of Janus MSi₂P_xAs_y (M = Mo, W) monolayers. *Phys. Rev. B* **2022**, *106*, No. 035417.
- (58) Wang, J.; Yip, S.; Phillpot, S.; Wolf, D. Crystal instabilities at finite strain. *Phys. Rev. Lett.* **1993**, *71*, 4182.
- (59) Zhang, T.; Ma, Y.; Yu, L.; Huang, B.; Dai, Y. Direction-control of anisotropic electronic behaviors via ferroelasticity in two-dimensional α -MPI (M = Zr, Hf). *Mater. Horiz.* **2019**, *6*, 1930.
- (60) Peng, R.; Ma, Y.; He, Z.; Huang, B.; Kou, L.; Dai, Y. Single-layer Ag₂S: a two-dimensional bidirectional auxetic semiconductor. *Nano Lett.* **2019**, *19*, 1227.
- (61) Liu, C.; Gao, H.; Li, Y.; Wang, K.; Burton, L.; Ren, W. Manipulation of the Rashba effect in layered tellurides MTe (M = Ge, Sn, Pb). *J. Mater. Chem. C* **2020**, *8*, 5143.
- (62) Ma, Y.; Dai, Y.; Guo, M.; Niu, C.; Zhu, Y.; Huang, B. Evidence of the existence of magnetism in pristine VX₂ monolayers (X = S, Se) and their strain-induced tunable magnetic properties. *ACS Nano* **2012**, *6*, 1695–1701.
- (63) Zhang, S.; Xu, R.; Duan, W.; Zou, X. Intrinsic Half-metallicity in 2D ternary chalcogenides with high critical temperature and controllable magnetization direction. *Adv. Funct. Mater.* **2019**, *29*, 1808380.
- (64) Amoroso, D.; Barone, P.; Picozzi, S. Spontaneous skyrmionic lattice from anisotropic symmetric exchange in a Ni-halide monolayer. *Nat. Commun.* **2020**, *11*, 5784.
- (65) Mogulkoc, A.; Modarresi, M.; Rudenko, A. Two-dimensional chromium pnictides CrX (X = P, As, Sb): half-metallic ferromagnets with high Curie temperature. *Phys. Rev. B* **2020**, *102*, No. 024441.
- (66) Shen, S.; Xu, X.; Huang, B.; Kou, L.; Dai, Y.; Ma, Y. Intrinsic triferroicity in a two-dimensional lattice. *Phys. Rev. B* **2021**, *103*, 144101.
- (67) Okubo, T.; Chung, S.; Kawamura, H. Multiple-q states and the skyrmion lattice of the triangular-lattice Heisenberg antiferromagnet under magnetic fields. *Phys. Rev. Lett.* **2012**, *108*, No. 017206.
- (68) Moriya, T. Anisotropic superexchange interaction and weak ferromagnetism. *Phys. Rev.* **1960**, *120*, 91–98.

(69) Liang, J.; Wang, W.; Du, H.; Hallal, A.; Garcia, K.; Chshiev, M.; Fert, A.; Yang, H. Very large Dzyaloshinskii-Moriya interaction in two-dimensional Janus manganese dichalcogenides and its application to realize skyrmion states. *Phys. Rev. B* **2020**, *101*, 184401.

(70) Dou, K.; Du, W.; Dai, Y.; Huang, B.; Ma, Y. Two-dimensional magnetoelectric multiferroics in a MnSTe/In₂Se₃ heterobilayer with ferroelectrically controllable skyrmions. *Phys. Rev. B* **2022**, *105*, 205427.

(71) Du, W.; Dou, K.; He, Z.; Dai, Y.; Huang, B.; Ma, Y. Spontaneous magnetic skyrmions in single-layer CrInX₃ (X = Te, Se). *Nano Lett.* **2022**, *22*, 3440–3446.

(72) Zhang, Y.; Xu, C.; Chen, P.; Nahas, Y.; Prokhorenko, S.; Bellaiche, L. Emergence of skyrmionium in a two-dimensional CrGe(Se,Te)₃ Janus monolayer. *Phys. Rev. B* **2020**, *102*, 241107.

(73) Grimme, S.; Antony, J.; Ehrlich, S.; Krieg, K. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *J. Chem. Phys.* **2010**, *132*, 154104.

(74) Kresse, G.; Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **1996**, *54*, 11169–11186.

(75) Blochl, P. Projector augmented-wave method. *Phys. Rev. B Condens Matter* **1994**, *50*, 17953–17979.

(76) Perdew, J.; Burke, K.; Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **1996**, *77*, 3865–3868.

(77) Anisimov, V.; Aryasetiawan, F.; Lichtenstein, A. First-principles calculations of the electronic structure and spectra of strongly correlated systems: the LDA+U method. *J. Phys.: Condens. Matter* **1997**, *9*, 767–808.

(78) Giannozzi, P.; Baroni, S.; Bonini, N.; Calandra, M.; Car, R.; Cavazzoni, C.; Ceresoli, D.; Chiarotti, G.; Cococcioni, M.; Dabo, I. QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials. *J. Phys.: Condens. Matter* **2009**, *21*, 395502.

(79) Miyatake, Y.; Yamamoto, M.; Kim, J.; Toyonaga, M.; Nagai, O. On the implementation of the ‘heat bath’ algorithms for Monte Carlo simulations of classical Heisenberg spin systems. *J. Phys. C: Solid State Phys.* **1986**, *19*, 2539–2546.

(80) Dou, K.; Du, W.; He, Z.; Dai, Y.; Huang, B.; Ma, Y. Antiferromagnetic skyrmion crystal in Janus monolayer CrSi₂N₂As₂. 2204.08792. arXiv. <https://arxiv.org/abs/2204.08792> (April 19, 2022).



CAS INSIGHTS™

EXPLORE THE INNOVATIONS SHAPING TOMORROW

Discover the latest scientific research and trends with CAS Insights. Subscribe for email updates on new articles, reports, and webinars at the intersection of science and innovation.

Subscribe today

CAS
A division of the American Chemical Society